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Mechatronic Sensors Trainer

Fundamental Labs – Thermocouple

V1.0 – 16th January 2026

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Caution

This equipment is designed to be used for educational and research purposes and is not intended for use by the public. The user is responsible for ensuring that the equipment will be used by technically qualified personnel only. Users are responsible for certifying any modifications or additions they make to the default configuration.

Mechatronic Sensors Trainer – Application Guide

Thermocouple Lab

Why Explore Thermocouple?

Thermocouples are essential temperature sensors used across engineering applications due to their wide operating range, low cost, accuracy and durability. They function based on the Seebeck effect, where a temperature difference between two different metals at a junction produces a measurable voltage. Thermocouples respond quickly to temperature changes and perform reliably in different environments. Understanding how they work and how to interpret their output is important since they are used for accurate temperature measurement and control in systems such as HVAC equipment, engines, furnaces, and industrial process controls.

Background

This lab is part of the Fundamentals labs for the Mechatronic Sensors Trainer. These labs will guide you through the sensors that are part of the board and the ones that are provided as external sensors so you can understand how to read from the sensors, what they read, when they are used, and you feel comfortable using them. This will also include highlighting datasheets to learn how to interpret them and know what to look for.

Prior to starting this lab, please review the following concept review (located in Documents/Quanser/4_concept_reviews/),

- 4_concept_reviews/Mechatronics/**019_thermocouple**.

Getting Started

In this lab, you will use the **Sensors Trainer** to learn about the basics of the **Thermocouples**. The lab will guide you through understanding how thermocouples work, how to read and use them as well as using the datasheet to understand the sensor better.

Before you begin this lab, ensure that the following criteria are met:

- Make sure you have a thermocouple, if not using the provided one, confirm it uses the same plug. Note the – and + on the sensor when plugging it into the device.
- Make sure the Sensors Trainer has been setup and tested. See the Quick Start Guide (Quanser/2_quick_start_guides/sensors_trainer) for details on this step.
- You are familiar with the basics of the programming language you are using for this lab.
- Have your development environment ready,
    - If using Python, we recommend using Visual Studio Code.